A Reproducible Evidence-Audit Framework for Leakage, Feature Stability, and Transportability in Translational Omics Classification

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Abstract

Objective: High-dimensional omics classifiers can appear credible when leakage, unstable feature selection, class imbalance, and weak transportability remain hidden. We developed the Reproducible Omics Evidence Audit (ROEA), a reporting framework for evaluating the evidence supporting such claims.

Methods: We contrasted an intentionally naive linear SVM workflow (global feature selection, fixed C = 1, and 5 × 5 repeated cross-validation) with a guarded nested workflow (training-fold feature selection and preprocessing, inner-loop C tuning, and five outer folds) in GSE25055, then projected a frozen discovery model to GSE25065 and GSE41998. The audit combined discrimination and operating-point metrics, bootstrap uncertainty, 1000 label permutations, 30 resampling seeds, feature stability, and external transportability.

Results: On real GSE25055 labels, AUROC was 0.7830 for the naive workflow and 0.7032 for the guarded workflow (contrast +0.0799, 95% CI +0.0080 to +0.1514). With fixed ROC direction, the naive permutation null remained strongly above chance (mean 0.8778), whereas the guarded null centred near chance (mean 0.4917). Feature stability was moderate (Nogueira 0.5128). External AUROCs were 0.5633 (GSE25065) and 0.6922 (GSE41998), with markedly cohort- and scaling-dependent operating-point behavior.

Conclusion: ROEA integrates established validation, permutation, stability, and external-validation methods into a reusable credibility audit. Because the compared workflows also

differed in tuning and resampling, their observed performance contrast is descriptive and should not be interpreted as the causal effect of feature-selection leakage alone.

Keywords: data leakage; transcriptomics; omics evidence audit; support vector machine; nested cross-validation; feature selection; feature stability

1. Introduction

1.1 Reproducibility challenge and study objective

Public transcriptomic repositories have enabled many machine-learning studies of cancer diagnosis, prognosis, treatment response, and biomarker discovery, letting historical expression data be reused to test computational hypotheses cheaply. This creates a difficult validation setting: thousands to tens of thousands of genes or probes are modeled using tens to hundreds of patient samples, so overfitting, unstable feature selection, and validation bias become central methodological risks rather than minor implementation details [1–3].

These risks matter especially when a study reports a gene signature as a candidate biomarker: a classifier may achieve strong internal performance while selecting different genes as training data are perturbed, weakening biological interpretability. Prior molecular-signature research shows feature-selection methods influence not only accuracy but also stability and interpretability [3], so transcriptomic classification studies require evidence about validation rigor, feature reproducibility, and external transportability — not just a single discrimination metric.

This manuscript is a methodological audit study using public breast cancer pathological complete response versus residual disease (pCR/RD) transcriptomic cohorts as a deliberately challenging stress test combining small sample size, high dimensionality, class imbalance, supervised feature selection, and external-cohort transportability constraints. The GEO cohorts GSE25055, GSE25065, and GSE41998 serve respectively as the discovery/internal-validation, same-study-family external-validation, and cross-platform transportability testbeds (Section 3.2).

1.2 Study objective and contributions

The objective was to develop and demonstrate a reproducible audit framework for data leakage, feature-selection stability, and external-cohort transportability in public translational omics classification.

The study makes five contributions: it contrasts an intentionally naive global-selection workflow with a guarded nested workflow; reports their performance contrast with uncertainty and label-shuffle negative controls; treats feature stability as a primary audit dimension; evaluates a frozen model on same-study-family and cross-platform cohorts; and consolidates these results into a reusable evidence-audit format. Because the workflows also differ in tuning and resampling, the comparison is descriptive rather than a one-factor estimate of leakage.

The intended contribution is ROEA, a reusable evidence-audit architecture for omics classifiers. This is not a new SVM algorithm, a clinically deployable prediction model, or a biomarker-discovery claim.

Statement of Significance

Problem. High-dimensional omics classifiers can appear credible when leakage, feature instability, class-imbalance behavior, and weak external transportability remain hidden.

Known context. Nested validation, leakage control, permutation controls, external validation, and stability metrics are established but are often reported separately.

Contribution. ROEA combines these elements in a measured evidence-audit format. In this study, the naive-versus-guarded result is a descriptive whole-workflow contrast because the arms also differ in tuning and resampling.

Relevance. The framework is intended for researchers, reviewers, editors, and translational teams evaluating high-dimensional prediction claims.

2. Related Work

2.1 Leakage-aware validation in high-dimensional biomedical modeling

A recurring concern in high-dimensional biomedical prediction is that validation estimates become biased when information from nominally held-out samples enters model development. In transcriptomic classification, this risk is amplified because the number of measured probes greatly exceeds the number of patients, making feature selection itself a major source of overfitting. Ambroise and McLachlan [1] showed that selecting genes on the full dataset before cross-validation produces strongly optimistic error estimates, and subsequent work frames this as a general reproducibility problem affecting any transformation, selection, tuning, or calibration step that learns from all samples before validation [4,5]. Independent reanalyses and reviews have documented such failures in molecular classifiers [24–27]. More recent work emphasizes genomics-specific machine-learning pitfalls [28], limited validation and transportability in gene-expression classifiers [29], and widespread pre-cross-validation leakage in cancer drug-response prediction [30].

Leakage control must therefore be embedded in the full validation architecture, not just the final classifier: scaling, imputation, filtering, class-imbalance handling, probability estimation, hyperparameter tuning, threshold selection, and feature selection can each become leakage pathways if fit outside the training partition. Nested cross-validation addresses this by separating inner-loop model selection from outer-loop performance estimation [6,7], and software such as nestedcv operationalises these principles for transcriptomic workflows [8].

The literature also leaves open how best to quantify leakage. A leaky-versus-guarded comparison identifies the effect of leakage only when the arms are matched on folds, repeats, tuning, and every other modeling choice. The present comparison does not meet that one-factor condition and is therefore interpreted as a whole-workflow contrast.

2.2 Feature-selection stability and interpretability of omics signatures

Leakage-free validation is necessary but not sufficient for credible omics biomarker claims: a classifier evaluated under an appropriate resampling design may still produce unstable feature lists that change substantially across folds or training-data perturbations. This matters because transcriptomic classifiers are often interpreted biologically through their selected genes, and prior work shows feature-selection methods affect not only predictive accuracy but also signature stability and interpretability [3]. Stability measures, including the framework of Nogueira et al. [9] and software such as stabm [10], quantify whether a selected feature set is reproducible rather than merely predictive in one split.

2.3 From model-building tools and checklists to evidence-audit reporting

Existing tools and guidance already provide many components needed for rigorous omics prediction studies. Linear support vector machines remain a useful baseline for high-dimensional gene-expression classification, operating in feature spaces much larger than the sample size and with a long history in cancer classification [11]. Omics-oriented packages such as nestedcv, stabm, and OmicSelector support leakage-resistant modelling, stability assessment, and automated feature-selection workflows [8,10,12].

A parallel body of work provides reporting guidance: REFORMS offers 32 consensus-based recommendations for machine-learning-based science with explicit attention to leakage [13], TRIPOD+AI specifies transparent reporting expectations for clinical prediction models [14], PROBAST+AI provides a framework for assessing prediction-model quality, risk of bias, and applicability [15], and Bernett et al. [5] supply diagnostic questions targeted at biological data.

The gap addressed here is not the absence of algorithms, software, or checklists, but that existing instruments commonly ask whether a safeguard was applied, whereas classifier credibility also turns on measured quantities: each workflow's behavior under randomized labels, cross-fold feature reproducibility, performance under class imbalance, and loss of performance when a frozen model meets an independent cohort (Section 5.4).

3. Methods

3.1 Study design overview

This public-data computational methodology study compared two linear SVM workflow families. The naive arm intentionally used global supervised feature selection, fixed C = 1, and 5 × 5 repeated cross-validation. The guarded arm confined feature selection, scaling, class weighting, and cost tuning to training data and used five outer folds with five-fold inner tuning. GSE25055 supported internal analysis; GSE25065 and GSE41998 supported frozen-model external assessment. Figure 1 contrasts the architectures. Because the arms differ in more than feature-selection placement, the comparison audits whole workflows rather than a matched ablation.

3.2 Public datasets and cohort eligibility

Eligible datasets were required to contain pretreatment bulk transcriptomic expression profiles, interpretable pCR/RD labels, adequate sample metadata, and public access through GEO or a comparable repository. Cohorts were excluded when response labels were ambiguous, samples overlapped with another validation cohort, the treatment time point was incompatible with pretreatment prediction, or platform mapping was insufficient for external projection.

- **Discovery / nested-CV cohort:** GSE25055 [16], a cohort of patients with invasive breast cancer treated with neoadjuvant taxane-anthracycline chemotherapy (Hatzis et al. 2011), with 306 usable labelled samples after excluding 4 samples with missing response labels.
- **External validation cohort 1:** GSE25065, the non-overlapping validation subseries from the same Hatzis et al. [16] study, with 182 usable labelled samples after excluding 16 samples with missing response labels.
- **Cross-platform sensitivity cohort:** GSE41998 (GPL571; [17]), from Horak et al. (2013) involving early-stage breast cancer patients treated with neoadjuvant doxorubicin/cyclophosphamide combined with ixabepilone or paclitaxel in a Phase II trial, with 253 evaluable samples.
- **Documented but not used:** GSE20194 and GSE20271, owing to MDACC-lineage patient-overlap risk.

3.3 Outcome definition

The primary endpoint was pathological complete response versus residual invasive disease. Samples with missing, ambiguous, post-treatment-only, or non-comparable outcome labels were excluded before modelling. In GSE41998, a live audit of the GEO Series Matrix confirmed 69 explicit Yes labels, 184 explicit No labels, 20 literal “0” values, and 6 missing values. Because the 20 zeros remained a distinct category in the parallel pcrrcb1 field and were not equivalent to explicit No, they and the 6 missing values were excluded before model evaluation. Full harmonisation decisions are provided in Supplementary File S1.

3.4 Data preprocessing and leakage-control rules

The central leakage-control rule was that any parameter estimated from data was fit on training data only and then applied unchanged to validation data. This covered scaling, filtering, class weighting, feature selection, hyperparameter tuning, thresholding, and probability estimation.

External cohorts were not used for feature selection, threshold adjustment, hyperparameter choice, or batch correction.

Author-processed GEO Series Matrix files were used without raw CEL reprocessing. Probe-level features were retained, near-zero-variance probes were removed, and z-scoring was fit within training partitions only. Labels were harmonised to pCR versus RD, excluding missing or ambiguous entries before modelling. Samples were sorted lexicographically by GEO sample accession before seeded folds were generated, making fold assignment independent of download order. The GPL96 (HG-U133A) platform provided 22,283 probe-level features; the near-zero-variance filter (variance cutoff 1×10^{-8}) removed no probes, so all 22,283 remained eligible for supervised t-test feature selection in GSE25055.

3.5 Pipeline A: naive/leaky baseline

The naive pipeline represents the flawed workflow this paper audits: t-test top-100 feature selection was performed once on the entire discovery cohort before any cross-validation split, after which a linear SVM (fixed cost = 1) was evaluated by 5 × 5 repeated cross-validation. It is not intended for valid inference; it quantifies the inflation a leaky analysis can produce.

3.6 Pipeline B: guarded nested linear SVM framework

The guarded pipeline used an outer loop for performance estimation and an inner loop for model selection. Within each outer split, all preprocessing, feature selection, class weighting, probability estimation, and SVM cost tuning were confined to the outer-training data; the outer test fold was evaluated once after the selected configuration was refit on the full outer-training set. The inner loop evaluated $C \in \{0.25, 1, 4\}$ by mean balanced accuracy. Performance was estimated with 5-fold outer cross-validation and model selection with 5-fold inner cross-validation, with folds stratified by pCR/RD status.

The workflows shared the same linear SVM, class-weighting rule, probability-estimation method, and top-100 t-test selector, but differed in feature-selection scope, cost policy, nesting, folds, and repetition, as summarised in Figure 1.

3.7 Model specification (linear SVM)

The primary classifier was a sequential-minimal-optimization-trained linear SVM [18], chosen as an interpretable workhorse for high-dimensional transcriptomic classification rather than a novel algorithm. Class weighting addressed pCR/RD imbalance and was recomputed within training partitions only ($w = N / (K \times n_c)$ for class c; balanced inverse-frequency weighting). SVM class-probability estimates were obtained by libsvm Platt scaling (e1071::svm, probability estimation enabled), fit within training partitions only; SMOTE [19,20] was not used.

3.8 Feature-selection methods

The reported empirical analysis used a univariate t-test top-K feature-selection strategy, with $K = 100$ in the primary analysis and $K \in \{25, 50, 200\}$ in a sensitivity sweep. Information gain or gain ratio, SVM recursive feature elimination, mRMR, CFS, and elastic-net embedded selection were not evaluated; Section 5.6 states the resulting scope limitation and Section 5.7 the planned extension.

3.9 Performance, leakage, and stability metrics

Classification performance was reported using AUROC, PR-AUC, balanced accuracy, sensitivity, specificity, and Matthews correlation coefficient (MCC) [21]. The naive-minus-guarded workflow contrast was calculated for each metric, with emphasis on ΔAUROC and ΔPR-AUC. Because the workflows differ in feature-selection scope, cost policy, and resampling design, this contrast is descriptive and is not a causal estimate of leakage alone. Feature stability was measured using the Nogueira stability index, feature-selection frequencies, and Jaccard overlap as an intuitive secondary metric.

For the guarded nested workflow, each metric was computed once on pooled out-of-fold predictions from the five outer folds. For the naive workflow, each metric was computed within each of five repeats and then averaged. Their difference is reported as a descriptive workflow contrast; the unequal repetition and fold structure preclude one-factor attribution.

Metric implementation. Because PR-AUC estimators differ, the implementation is stated explicitly: PR-AUC was computed by trapezoidal integration of the precision–recall curve (PRROC::pr.curve(…)$auc.integral). Recomputing the same predictions with step-wise average precision (as in scikit-learn) yielded values higher by approximately 0.006–0.011 across the four analysis settings. AUROC was computed with pROC using class levels RD then pCR and direction "<" fixed a priori, so larger pCR probabilities indicate pCR and ROC direction is not selected from the observed or permuted data. Table 5 therefore requires the PR-AUC estimator to be named.

3.10 Uncertainty quantification, permutation control, and repeated cross-validation

Uncertainty was quantified using stratified percentile bootstrap confidence intervals (B = 2000, stratified within the pCR and RD classes so that class prevalence is fixed) applied to stored out-of-fold or external predictions, without model refitting during resampling. The workflow contrast was evaluated as a paired naive-minus-guarded difference on the same resampled samples; the reported bootstrap p-value is two-sided, computed as $2 \times \min(\Pr[\Delta > 0], \Pr[\Delta < 0])$ over the bootstrap replicates.

As a secondary check, DeLong tests were computed per repeat (leaky repeat r vs. guarded nested), since the guarded pipeline yields one out-of-fold score per sample while the leaky baseline yields one per repeat, and collapsing repeats would not match the repeat-averaged estimate (Section 4.3).

A label-shuffle negative control reran both complete workflows on 1000 permuted pCR/RD label sets, with permutation preceding every modeling step and ROC direction fixed a priori. The naive arm's global feature selection was re-executed against each shuffled vector, so the null characterizes the behavior of each full workflow under randomized labels. Permutation p-values are Pr[null ≥ observed], computed as $(\#\{\text{null} \geq \text{observed}\} + 1)/(B + 1)$. Robustness to fold assignment was assessed across 30 matched random seeds; the seed used for the primary single-split analyses (20260620) is the anchor seed throughout.

3.11 External validation protocol

After discovery analysis, the final feature set, SVM cost, threshold, and scaling parameters were frozen before external projection: external labels were used only for final evaluation, with no feature re-selection, threshold tuning, model refitting, or supervised cross-cohort normalization performed.

The frozen model was constructed as follows. It was fit on the full GSE25055 discovery cohort using the prespecified top-100 t-test selector, with C = 4 selected by a discovery-only five-fold cross-validation on the full cohort that maximized mean balanced accuracy over the grid {0.25, 1, 4}. This full-cohort selection was made independently of the nested outer-fold cost choices, which varied across folds; the nested analysis estimates performance, whereas this single fitted model is the artifact projected externally. External cohorts entered no model-development step.

Cross-platform projection from GPL96 to GPL571 used a direct probe-identifier intersection (the exact GPL96 ∩ GPL571 overlap); all 100 frozen discovery probes were present on GPL571, and no outcome labels were used during platform mapping.

Operating-point metrics (sensitivity, specificity, balanced accuracy, MCC) were computed from the default class labels returned by the trained SVM (`e1071::predict`), and AUROC and PR-AUC from the corresponding libsvm probability scores. This decision rule was fixed on discovery data and applied unchanged to the external cohorts, with no Youden-index, balanced-accuracy, or external-cohort threshold optimization. The default rule was retained deliberately, since optimizing it would itself require fitting on data, reintroducing the tuning the audit is designed to detect. The consequence — uniformly low pCR sensitivity — is reported as an audit finding rather than corrected away; Section 5.6 states the resulting limitation.

3.12 Reproducibility and software availability

The analysis was designed for reproducibility by recording data-accession sources, preprocessing decisions, random seeds, fold assignments, selected-feature lists, software environment information, and audit tables. A prior project snapshot is Zenodo DOI 10.5281/zenodo.21134086. The synchronized v1.2.0 code, results, metadata, figures, supplements, and manuscript are archived through the Zenodo concept record at DOI 10.5281/zenodo.21134085. The corrected permutation analysis refits both complete workflows after every label shuffle. Supplementary File S4 reports the software environment, and Supplementary File S5 provides the evidence-audit dashboard.

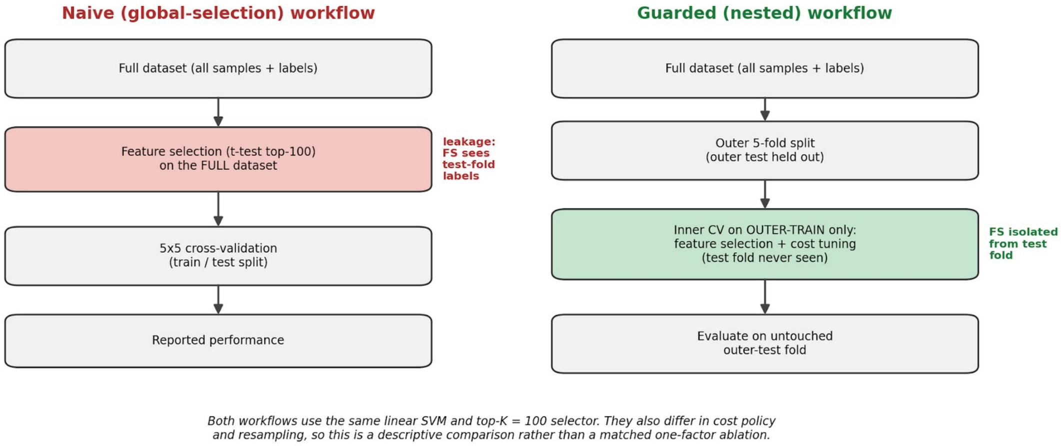

Figure 1. Conceptual comparison of the leaky (global) and guarded (nested) pipelines on GSE25055. The naive baseline intentionally performs supervised feature selection on the full dataset before flat 5 × 5 repeated cross-validation with fixed C = 1. The guarded workflow nests feature selection, scaling, class weighting, and C tuning inside training data before five-fold outer evaluation. Both use a linear SVM and top-100 t-test selector, but the contrast is not a matched one-factor ablation.

4. Results

The contribution reported here is methodological and audit-oriented: the analysis characterizes leakage sensitivity, feature reproducibility, class-imbalance behavior, and external-validation transportability for a leakage-aware linear SVM workflow. It does not claim clinical biomarker discovery.

4.1 Cohort construction and endpoint harmonization

After label harmonization, the usable sample counts were 306 for GSE25055, 182 for GSE25065, and 253 for GSE41998. GSE20194 and GSE20271 were documented but not analyzed because of patient-overlap risk. Cohort details are summarised in Table 1.

Table 1. Cohort construction and endpoint harmonization.

Cohort	Role/use	Usable labels	Audit note
GSE25055	Discovery / internal validation	306 (57 pCR, 249 RD)	GPL96; 4 missing labels excluded.
GSE25065	Same-platform, same-study-family validation	182 (42 pCR, 140 RD)	GPL96; 16 missing labels excluded; non-overlapping with GSE25055.
GSE41998	Cross-platform	253 evaluable	GPL571; 253 evaluable after

Cohort	Role/use	Usable labels	Audit note
	transportability sensitivity	(69 pCR, 184 RD)	excluding 20 undocumented "0" codes and 6 missing/unclear labels; exact GPL96 ∩ GPL571 projection; no retuning.
GSE20194	Sensitivity candidate	278 (56 pCR, 222 RD)	GPL96; held because of MDACC lineage overlap risk.
GSE20271	Sensitivity candidate	178 (26 pCR, 152 RD)	GPL96; held because of MDACC lineage overlap risk.

Note. GSE20194 and GSE20271 were documented for audit transparency but were not used. pCR = pathological complete response; RD = residual disease.

4.2 Naive versus guarded linear SVM workflow performance

On GSE25055 the naive workflow produced higher apparent threshold-independent discrimination than the guarded workflow: AUROC 0.7830 versus 0.7032 and PR-AUC 0.4257 versus 0.3476. The descriptive contrasts were +0.0799 for AUROC and +0.0781 for PR-AUC; operating-point metrics varied under class imbalance.

Table 2. Naive versus guarded linear SVM workflow performance on GSE25055.

Metric	Naive workflow	Guarded workflow	Naive − guarded contrast
AUROC	0.7830 (0.728–0.836)	0.7032 (0.628–0.772)	+0.0799 (95% CI +0.0080 to +0.1514; bootstrap p = 0.026)
PR-AUC	0.4257 (0.350–0.521)	0.3476 (0.267–0.467)	+0.0781 (−0.0423 to +0.1755; p = 0.192)
Balanced accuracy	0.5525 (0.525–0.582)	0.5724 (0.524–0.625)	−0.020
MCC	0.1982 (0.102–0.287)	0.2138 (0.076–0.350)	−0.016
Sensitivity (pCR)	0.1298 (0.077–0.190)	0.1930 (0.105–0.298)	−0.063
Specificity	0.9751 (0.963–0.986)	0.9518 (0.924–0.976)	+0.023

Note. Values are point estimates with 95% stratified-bootstrap confidence intervals (B = 2000). The AUROC contrast excluded zero, whereas the PR-AUC contrast did not. Both remain descriptive whole-workflow comparisons because the arms differ in tuning and resampling. PR-AUC was computed by trapezoidal integration (Section 3.9); MCC = Matthews correlation coefficient.

4.3 Uncertainty quantification for the workflow contrast

Bootstrap confidence intervals placed the real-label workflow contrast in context: ΔAUROC was +0.0799 (95% CI +0.0080 to +0.1514; p = 0.026), and ΔPR-AUC was +0.0781 (−0.0423 to +0.1755; p

= 0.192). Per-repeat DeLong comparisons gave AUROC differences of +0.0661 to +0.0901 (p = 0.020 –0.117), showing mixed repeat-level resolution.

These results do not quantify the causal effect of feature-selection leakage, because the two workflows also differ in tuning and resampling. The permutation analysis instead asks how each complete workflow behaves when the labels contain no predictive information.

4.4 Sensitivity to the feature-selection budget (top-K sweep)

Across K = 25, 50, 100, and 200 the naive workflow showed higher AUROC and PR-AUC than the guarded workflow at every budget, while feature-selection stability rose with larger K (Nogueira 0.3513 to 0.5752; full per-K estimates in Supplementary File S2, topK_sweep_sensitivity). This sensitivity analysis supports the direction of the whole-workflow contrast across feature budgets but does not isolate feature-selection placement or test other selectors or model classes.

4.5 Permutation negative control of workflow behavior

The label-shuffle control re-executed both complete workflows on 1000 permuted label vectors, repeating global feature selection in the naive arm against each shuffled vector. ROC direction was fixed before analysis.

The guarded workflow's AUROC null centred near chance (mean 0.4917, SD 0.0514, 2.5th–97.5th percentile 0.3902–0.5836), and its PR-AUC null centred at 0.1898 against a prevalence reference of 0.1863. Its real-label AUROC and PR-AUC exceeded their respective null distributions (permutation p = 0.001 for both), supporting non-random predictive structure in this dataset without establishing a biological mechanism or clinical utility.

The naive workflow behaved differently: across 1000 shuffled label vectors its mean AUROC was 0.8778 (SD 0.0512, 2.5th–97.5th percentile 0.7566–0.9575), with all null values above 0.5, and its mean PR-AUC was 0.6623. Its real-label AUROC of 0.7830 was not exceptional relative to its own null (p = 0.951). Because supervised feature selection used the complete shuffled-label vector before cross-validation, these results show that the naive workflow does not provide a valid out-of-sample null benchmark.

Under shuffled labels, the mean naive-minus-guarded workflow contrast was +0.3860 for AUROC (2.5th–97.5th percentile +0.2271 to +0.5209) and +0.4725 for PR-AUC, compared with +0.0799 and +0.0781 on real labels. Because the arms differ in feature-selection placement, tuning, folds, and repetition, neither the real-label nor shuffled-label contrast estimates the isolated causal effect of leakage. The permutation results instead characterize the behavior of the two complete workflows and motivate a future matched one-factor ablation (Table 3, Figure 2).

The guarded AUROC null was centred close to 0.5, with 43% of permutations above chance (Table 3), as expected when ROC orientation is fixed a priori.

Table 3. Permutation negative control, 1000 label shuffles, GSE25055.

Statistic	Observed	Chance reference	Null mean (SD)	Null 2.5th–97.5th pct	p (null ≥ observed)
Naive AUROC	0.7830	0.5000	0.8778 (0.0512)	0.7566–0.9575	0.951

Statistic	Observed	Chance reference	Null mean (SD)	Null 2.5th–97.5th pct	p (null ≥ observed)
Guarded AUROC	0.7032	0.5000	0.4917 (0.0514)	0.3902–0.5836	0.001
ΔAUROC (naive – guarded)	+0.0799	0.0000	+0.3860 (0.0753)	+0.2271 to +0.5209	1.000
Naive PR-AUC	0.4257	0.1863	0.6623 (0.1162)	0.4087–0.8592	0.966
Guarded PR-AUC	0.3476	0.1863	0.1898 (0.0293)	0.1465–0.2622	0.001
ΔPR-AUC (naive – guarded)	+0.0781	0.0000	+0.4725 (0.1215)	+0.2124 to +0.6819	1.000

Note. The chance reference is 0.5 for AUROC and the pCR prevalence (0.1863) for PR-AUC. A p-value near 1 indicates that the observed value is unexceptional or low relative to the complete workflow's own null distribution. Permutation was applied before all modeling steps, so global feature selection was re-executed on every shuffled label vector.

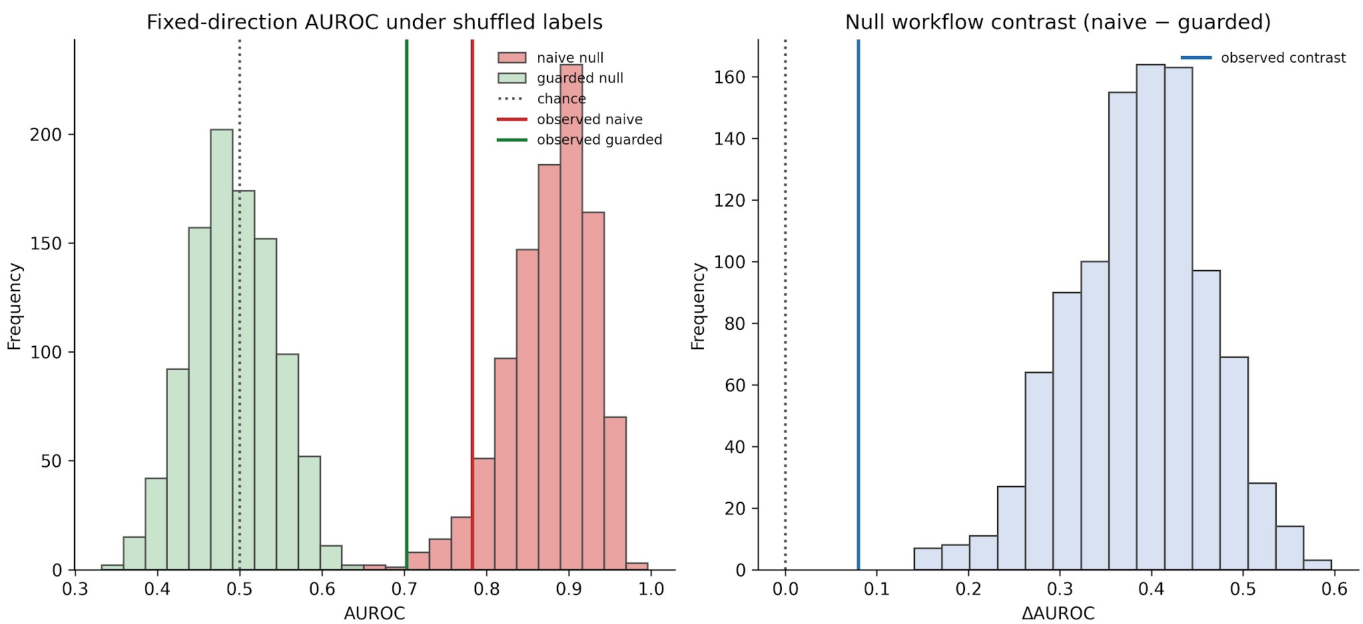

Figure 2. Fixed-direction AUROC null distributions from 1000 label permutations of the complete naive and guarded workflows on GSE25055. Vertical lines mark each workflow's observed real-label AUROC. Every modeling step was rerun after label shuffling.

4.6 Seed-level robustness across 30 repeated cross-validation runs

Across 30 matched random seeds, the real-label workflow contrast was positive in 28 of 30 seeds for AUROC and 22 of 30 seeds for PR-AUC. Median contrasts were +0.0397 and +0.0473, respectively. These seed-level analyses evaluate robustness to fold assignment, not patient-level inference or the causal effect of feature-selection placement; full outputs are provided in Supplementary File S2.

4.7 Feature-stability audit

The guarded nested pipeline showed moderate feature stability. The values in this section are from the anchor seed 20260620, the single split used for the primary audit; Supplementary File S2 gives the corresponding distributions across all 30 seeds.

Across the five outer folds of the anchor seed, 229 unique probes were selected at least once; 26 appeared in all five folds, and 105 appeared in only one. Mean Jaccard overlap was 0.3487, and the Nogueira stability index was 0.5128. The 30-seed medians were 29 core probes, 105 single-fold probes, Jaccard 0.3765, and Nogueira 0.5435. Figure 3 summarises the stable core and unstable tail. Selected-feature lists and fold assignments are provided in Supplementary File S3.

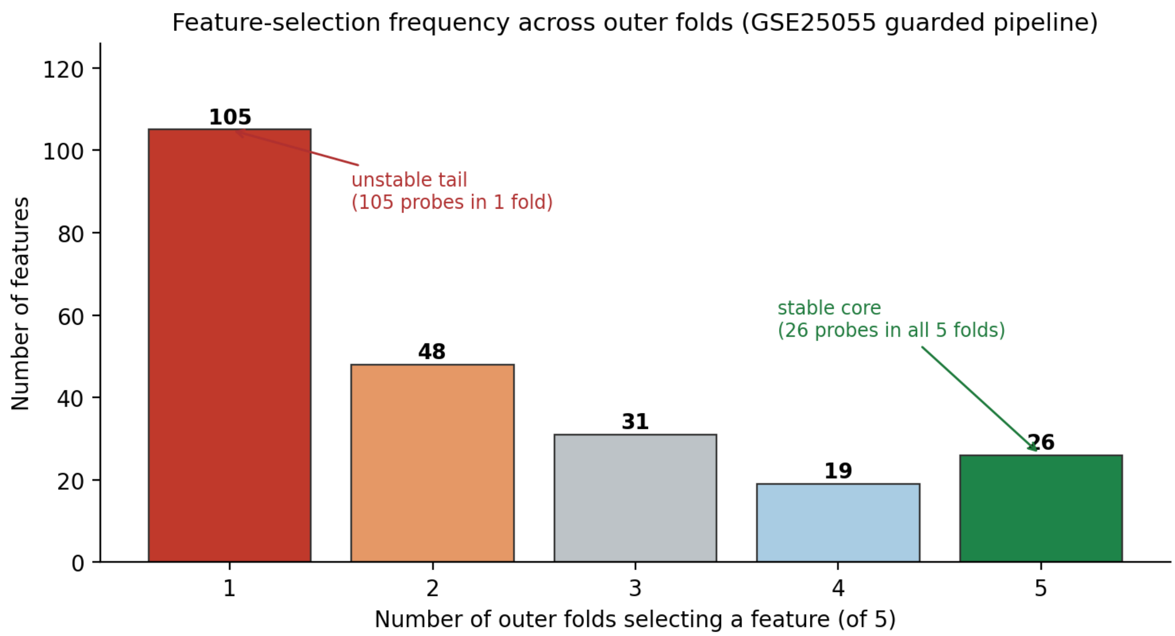

Figure 3. Feature-selection frequency across the five outer folds of the guarded workflow on GSE25055 (anchor seed 20260620). A stable core of 26 probes selected in all five folds coexists with an unstable tail of 105 probes selected in only one fold.

4.8 Same-platform, same-study-family validation on GSE25065

The frozen guarded model achieved AUROC 0.5633 and PR-AUC 0.2986 on GSE25065, with high specificity but low pCR sensitivity (5 true positives, 8 false positives, 132 true negatives, 37 false negatives). Relative to discovery nested validation, this represents an external transportability decline rather than a leakage effect. Results are integrated with the cross-platform cohort in Table 4.

4.9 Cross-platform transportability sensitivity on GSE41998

On the independent cross-platform cohort GSE41998, the same frozen GSE25055 model recovered all 100 probes and achieved AUROC 0.6922 and PR-AUC 0.4634 under discovery-derived scaling. At the fixed decision rule, sensitivity was 0.7971 and specificity was 0.5000, showing that discrimination and operating-point behavior conveyed different information.

Because cross-platform projection requires a scaling decision, a prespecified sensitivity analysis repeated it using within-cohort z-scoring instead of frozen discovery-derived scaling (AUROC 0.7067, PR-AUC 0.4674, balanced accuracy 0.5317, MCC 0.1235, sensitivity 0.1014, specificity 0.9620). Threshold-independent discrimination was similar across scaling choices (AUROC 0.692 vs 0.707), but operating-point behavior reversed sharply, with sensitivity falling from 0.797 to 0.101 as specificity rose from 0.500 to 0.962. Frozen discovery-derived scaling is primary because it uses no external-cohort estimates; within-cohort z-scoring is a sensitivity analysis, not a tuning opportunity.

Table 4. External transportability of the frozen GSE25055 model.

Metric	Discovery guarded CV (GSE25055)	GSE25065 (same-study-family)	GSE41998 (cross-platform sensitivity)
AUROC	0.7032	0.5633 (0.463–0.661)	0.6922 (0.619–0.766)
PR-AUC	0.3476	0.2986 (0.222–0.414)	0.4634 (0.371–0.570)
Balanced accuracy	0.5724	0.5310 (0.482–0.588)	0.6486 (0.590–0.710)
MCC	0.2138	0.1013 (−0.060–0.272)	0.2682 (0.161–0.376)
Sensitivity (pCR)	0.1930	0.1190 (0.024–0.215)	0.7971 (0.696–0.884)
Specificity	0.9518	0.9429 (0.900–0.979)	0.5000 (0.429–0.571)

Note. Both external cohorts were projected once, with no refitting, retuning, threshold adjustment, or feature re-selection. Values in the external columns include 95% stratified percentile bootstrap confidence intervals (B = 2000) computed from stored frozen-model predictions, with no model refitting during resampling. For GSE41998, the frozen model recovered 100 of 100 probes (z-scoring sensitivity variant: Section 4.9).

4.10 Integrated Reproducible Omics Evidence Audit

The analyses were consolidated into the Reproducible Omics Evidence Audit (ROEA, Table 5), covering dataset integrity, leakage sensitivity, guarded performance, feature stability, external transportability, class-imbalance behavior, reproducibility status, and unresolved risks. The domains are designed for reuse across biomedical modeling studies, but this empirical demonstration is limited to one classifier and one selector family (Section 5.6).

Table 5. Reproducible Omics Evidence Audit (ROEA): a reusable reporting instrument.

Audit domain	What to report and how to derive it	Reporting standard / red-flag trigger	This study (worked example)
Dataset integrity	Usable N and class	Report N and class	Discovery 306 (57

Audit domain	What to report and how to derive it	Reporting standard / red-flag trigger	This study (worked example)
	balance per cohort, platform (GPL), and all excluded/held datasets, derived by reconciling each accession to analysis-ready labels and documenting exclusions and patient- or lineage-overlap risk.	counts per cohort; disclose held datasets. Flag undocumented provenance, exclusions, or overlap risk.	pCR, 249 RD), GPL96; GSE20194 and GSE20271 held for MDACC-lineage overlap.
Leakage sensitivity	Documented naive-versus-guarded contrast and empirical null for each workflow, with all modeling steps re-executed after label shuffling. Use matched folds, repeats, tuning, and preprocessing when the goal is causal attribution to a single leakage pathway.	Report each workflow's permutation null and the complete resampling design. Flag automatically oriented AUROC, an above-chance null, or causal attribution from unmatched workflow arms.	Real-label ΔAUROC +0.0799 (95% CI +0.0080 to +0.1514); null-label ΔAUROC +0.3860. Naive observed AUROC 0.7830 was unexceptional relative to its null mean 0.8778 ($p = 0.951$); guarded observed 0.7032 exceeded its null mean 0.4917 ($p = 0.001$).
Guarded (nested) validation performance	Threshold-independent and operating-point metrics from the leakage-controlled pipeline (AUROC, PR-AUC, balanced accuracy, MCC), with the estimator implementation named.	Report PR-AUC and MCC alongside AUROC, and name the PR-AUC estimator. Flag reliance on AUROC alone under class imbalance, or an unnamed PR-AUC implementation.	AUROC 0.7032; PR-AUC 0.3476 (trapezoidal integration; step-wise average precision 0.3590); balanced accuracy 0.5724; MCC 0.2138.
Feature stability	Fold-level selection stability (Nogueira index, mean Jaccard) and the stable-core versus unstable-tail profile, with the seed or split identified.	Report stability alongside any gene list. Flag a single reported gene list with no fold-level stability evidence.	Anchor seed: Nogueira 0.5128, mean Jaccard 0.3487; 26 core probes versus 105 single-fold probes; 30-seed medians 0.5435, 0.3765, 29, and 105.
External transportability	Frozen-model performance on independent same-platform and cross-	Report the external decline explicitly. Flag any retuning, threshold adjustment, or feature	GSE25065 AUROC 0.5633; GSE41998 AUROC 0.6922 (0.7067 under within-cohort

Audit domain	What to report and how to derive it	Reporting standard / red-flag trigger	This study (worked example)
	platform cohorts, plus scaling/mapping sensitivity, derived by freezing features, scaling, threshold, and hyperparameters on discovery and projecting once.	re-selection on external data.	scaling), frozen model projected once.
Class-imbalance behavior	Minority-class sensitivity and specificity at the operating point for each cohort, and how the operating point was chosen.	Flag high specificity paired with low minority-class sensitivity presented as strong performance, and any threshold selected on the evaluation cohort.	GSE25065 sensitivity 0.119 / specificity 0.943. In GSE41998, scaling changed sensitivity/specificity from 0.797/0.500 to 0.101/0.962; the decision rule was never re-tuned.
Reproducibility status	Availability of code, data-accession lists, random seeds, fold assignments, selected-feature outputs, and result-generating scripts, assessed by whether a version-tagged, DOI-citable release can regenerate the reported analyses.	Flag results that cannot be regenerated from an archived artifact.	Submission-aligned v1.2.0 is archived through Zenodo concept DOI 10.5281/zenodo.21134085; DOI 10.5281/zenodo.21134086 is a prior snapshot.
Limitations / unresolved risks	Explicit statement of validation scope, unresolved risks, and biomarker-claim status.	Flag any biomarker claim not backed by prospective or mechanistic validation.	Single discovery cohort; one model class and selector family; unmatched workflow arms; PR-AUC contrast CI includes zero; no clinical biomarker claim — recurrent probes are audit outputs, not validated biomarkers.

Note. This is a reusable reporting instrument, not a results summary; the same domains can be populated for other omics platforms, endpoints, selectors, and model classes.

5. Discussion

5.1 Principal findings

Four findings calibrate the workflow comparison. First, the real-label AUROC contrast was +0.0799 (95% CI +0.0080 to +0.1514), whereas the PR-AUC contrast remained uncertain. Second, with ROC direction fixed a priori, the naive workflow's null AUROC averaged 0.8778 under randomized labels, while the guarded null averaged 0.4917 and its real-label result exceeded that null ($p = 0.001$). Third, the direction of the descriptive contrast persisted across most seeds, although the arms were not matched one factor at a time. Finally, external projection revealed transportability and operating-point limitations distinct from internal validation bias.

These findings support leakage-aware, uncertainty-bounded auditing over reliance on a single headline AUROC [22]. Interpreting the null behavior of each complete workflow, together with stability and external validation, is more informative than assigning a causal meaning to an unmatched between-workflow difference [23].

5.2 Why leakage-aware validation changes omics signature interpretation

The permutation results expose the failure mode of global supervised feature selection: information about samples later used for testing enters the selected feature set. With 22,283 candidate probes and 306 samples, the naive top-100 workflow produced strongly above-chance apparent discrimination even for randomized labels. Its real-label AUROC was unexceptional relative to that workflow-specific null, so the naive estimate cannot support a biological or clinical interpretation.

The reporting implication is to present the null distribution of each workflow, not only their difference. The shuffled-label contrast (+0.3860 AUROC) was larger than the real-label contrast (+0.0799), but it remains a comparison of complete workflows because the arms also differ in tuning and resampling. A matched ablation is required before assigning an effect size to feature-selection placement.

The guarded workflow's fixed-direction null centred close to chance (Table 3). Reporting that empirical null remains valuable because it verifies the behavior of the implemented workflow rather than assuming that a design label alone guarantees unbiased execution.

5.3 Feature stability versus predictive performance

The stability audit shows that predictive performance and feature reproducibility answer different questions: the guarded pipeline selected a stable core across all folds but also a large unstable tail selected only once, so transcriptomic machine-learning papers should not report a single gene list without fold-level stability evidence. These features should be interpreted as audit outputs, not validated biomarkers; repeated-seed analysis confirmed the moderate stability was reproducible across seeds, reinforcing that the audit characterizes the selection procedure rather than a fixed gene panel.

5.4 Relation to existing tools, checklists, and literature

This work complements nestedcv [8], stabm [10], and OmicSelector [12]. ROEA focuses on the quantities that should be measured and the patterns that should raise concern, rather than replacing model-building or stability software.

5.5 Practical recommendations

1. Never perform supervised feature selection before splitting.
2. Fit scaling and preprocessing parameters on training data only.
3. Perform feature selection and hyperparameter tuning inside nested cross-validation.
4. If resampling or class-balancing (SMOTE, over/under-sampling, class weighting) is used, fit it inside training folds only, as done here with class weighting (no SMOTE).
5. Report PR-AUC, MCC, balanced accuracy, sensitivity, and specificity for imbalanced endpoints, and name the PR-AUC estimator, since implementations can differ by an amount comparable to the effects reported.
6. Report a permutation null for every workflow compared, not just their difference, and treat a lower-tail observed value as a warning of biased workflow behavior.
7. Report feature stability and feature-selection frequencies in biomarker studies, identifying the seed or split.
8. Freeze features, thresholds, and hyperparameters before external validation, and report scaling-choice sensitivity for cross-platform projection.
9. Release code, data-accession lists, random seeds, fold assignments, and selected-feature outputs.

5.6 Limitations

The discovery analysis rests on one cohort. The AUROC workflow-contrast confidence interval excluded zero, whereas the PR-AUC interval did not. More importantly, the naive and guarded arms differ in feature-selection scope, cost policy, nesting, folds, and repetition; their difference therefore cannot isolate the causal effect of feature-selection leakage.

The audit domains are intended to be model- and selector-neutral, but the worked example uses only a linear SVM and t-test top-K selection. Generalization to embedded selectors, nonlinear models, and other outcomes remains untested. A matched ablation that holds folds, repeats, tuning, and all preprocessing constant except feature-selection placement is also needed to estimate the effect of that leakage pathway.

The 30-seed analysis addresses fold-assignment robustness, not patient-level inference; external validation was limited to one same-study-family and one cross-platform cohort. Operating-point behavior was poor or unstable: GSE25065 had low pCR sensitivity, while GSE41998's sensitivity-specificity trade-off changed sharply with scaling. Because the decision rule was not optimized externally, these results describe the frozen artifact rather than the best achievable clinical trade-off. GSE20194 and GSE20271 were held for patient-overlap risk; calibration was not assessed, and no prospective clinical, mechanistic, or wet-lab validation was performed. Recurrent probes are audit outputs, not validated biomarkers.

5.7 Future work

The immediate extensions are a matched one-factor leakage ablation and a second modeling family, such as elastic-net logistic regression, to test whether the observed null behavior and stability profile generalize beyond filter-based linear SVMs. Further work could address RNA-seq, multi-omics, survival endpoints, calibration, other cancers, and broader multi-cohort validation.

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6. Conclusion

In high-dimensional translational omics classification, apparent performance can be distorted when supervised feature selection and model development escape strict validation separation. In GSE25055, the naive workflow reported higher AUROC and PR-AUC than the guarded nested workflow, while its fixed-direction permutation null remained strongly above chance; the guarded null centred near 0.5. These results identify invalid null behavior in the naive workflow. Because the two arms also differ in tuning and resampling, their contrasts are descriptive and do not estimate the causal magnitude of feature-selection leakage.

The Reproducible Omics Evidence Audit (Table 5) is proposed as a standardized communication format for high-dimensional biomedical prediction studies. Its domains — dataset integrity, leakage sensitivity, guarded validation performance, feature stability, external transportability, class-imbalance behavior, reproducibility status, and unresolved risks — are specified independently of breast cancer, pCR prediction, microarrays, and SVMs, and could extend to RNA-seq, multi-omics, imaging-derived features, EHR phenotypes, survival endpoints, alternative selectors, and other model classes; this study demonstrates the instrument for one model class only. Used prospectively, such an audit can help reviewers, editors, and investigators distinguish evidence that survives appropriate controls from validation artifacts and compare omics-classifier claims transparently.

List of Abbreviations

AUROC, area under the receiver operating characteristic curve; GEO, Gene Expression Omnibus; MCC, Matthews correlation coefficient; NCV, nested cross-validation; pCR, pathological complete response; PR-AUC, area under the precision–recall curve; RD, residual disease; RFE, recursive feature elimination; ROEA, Reproducible Omics Evidence Audit; SMO, sequential minimal optimization; SMOTE, synthetic minority over-sampling technique; SVM, support vector machine.

Declarations

Ethics approval and consent to participate

This study uses publicly available, de-identified transcriptomic datasets, for which no new human participants were recruited, and no separate ethical approval was required.

Consent for publication

Not applicable.

Availability of data and materials

The datasets analyzed are publicly available from the NCBI Gene Expression Omnibus under accessions GSE25055, GSE25065, GSE41998, GSE20194, and GSE20271. Cohort roles and exclusion decisions are described in the Methods, Results, and Supplementary File S1. The submission-aligned analysis code, reproducibility materials, figures, tables, supplements, and manuscript are hosted in the public GitHub repository (https://github.com/iry496/leakage-aware-smo-svm-omics; code under the MIT license, manuscript-derived materials under CC-BY-4.0) and archived through the Zenodo concept record at DOI 10.5281/zenodo.21134085. No new data were generated by this study.

Software availability

Project name: Reproducible Omics Evidence Audit. Project home page: https://github.com/iry496/leakage-aware-smo-svm-omics. Versioned archive: Zenodo concept DOI 10.5281/zenodo.21134085. Programming language: R. License: MIT for code and CC-BY-4.0 for manuscript-derived materials. The executed software environment is reported in Supplementary File S4.

Supplementary material

Supplementary File S1. Dataset harmonization decisions and held-cohort audit. Supplementary File S2. Full 30-seed robustness outputs. Supplementary File S3. Selected-feature lists and fold assignments. Supplementary File S4. Reproducibility environment report. Supplementary File S5. Evidence-audit dashboard.

Competing interests

The authors declare that they have no competing interests.

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Author Contributions

Iris Yang: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Writing – Original Draft, Visualization, Project administration. Chung-I Huang: Methodology, Validation, Writing – Review & Editing. Paul Tan: Validation, Writing – Review & Editing. Jewel

Wang: Resources, Writing – Review & Editing. Jung Chen: Validation, Writing – Review & Editing. Wing-Keung Wong: Supervision, Methodology, Writing – Review & Editing.

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During manuscript preparation, the authors used OpenAI ChatGPT and Codex to support organization, language editing, document formatting, and pre-submission compliance checking. These tools were not used to generate or alter the study data, analyses, submitted figures, or graphical abstract. The authors verified the scientific claims, numerical results, citations, and final wording and take full responsibility for the manuscript.

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